

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

Inference from sample statistics and margin of error

02

1 questions · ~2 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

A bag containing 10,000 beads of assorted colors is purchased from a craft store. To estimate the percent of red beads in the bag, a sample of beads is selected at random. The percent of red beads in the bag was estimated to be 15%, with an associated margin of error of 2%. If r is the actual number of red beads in the bag, which of the following is most plausible?

- A. $r > 1,700$
- B. $1,300 < r < 1,700$
- C. $200 < r < 1,500$
- D. $r < 1,300$